

Procedure	Stages	Cpl	Loc	Blk	Op	Collective C sq S/W	Blocked resources and remarks
MPI_SEND	i-s-c-f	c+f	nl	b	b-op	-	
MPI_SSEND	i-s-c-f	c+f	nl	b	b-op	-	1)
MPI_RSEND	i-s-c-f	c+f	l	b	b-op	-	12) 17)
MPI_BSEND	i-s-c-f	c+f	l	b	b-op	-	12)
MPI_RECV	i-s-c-f	c+f	nl	b	b-op	-	
MPI_ISEND, MPI_ISSEND	i-s---	ic	l	nb	nb-op	-	buffer
MPI_IRECV	i-s---	ic	l	nb	nb-op	-	buffer
corresponding MPI_WAIT	---c-f	c+f	nl		nb-op	-	
corr. MPI_TEST returning flag=TRUE	---c-f	c+f	l		nb-op	-	
corr. MPI_TEST returning flag=FALSE	-----		l		nb-op	-	buffer cached on req
MPI_IBSEND, MPI_IRSEND	i-s---	ic	l	nb	nb-op	-	buffer
corresponding MPI_WAIT	---c-f	c+f	l		nb-op	-	3) 17)
corr. MPI_TEST returning flag=TRUE	---c-f	c+f	l		nb-op	-	
corr. MPI_TEST returning flag=FALSE	-----		l		nb-op	-	buffer 7)
MPI_PROBE	-----	c	nl			-	2)
MPI_IPROBE	-----	c	l			-	2) 11)
MPI_MPROBE	i-----	ic	nl	b	b-op	-	message 8)
MPI_IMPROBE	i-----	ic	l	nb	b-op	-	message 15)
MPI_MRECV of a probed message	i-s-c-f	c+f	l	b	b-op	-	13)
MPI_IMRECV of a probed message	i-s---	ic	l	nb	nb-op	-	buffer 13)
corresponding MPI_WAIT	---c-f	c+f	l		nb-op	-	
corr. MPI_TEST returning flag=TRUE	---c-f	c+f	l		nb-op	-	
corr. MPI_TEST returning flag=FALSE	-----		l		nb-op	-	buffer 7)
MPI_(-[S B R]SEND_INIT, MPI_RECV_INIT	i-----	ic	l	nb	p-op	-	buffer address 9) 14)
corresponding MPI_START, MPI_STARTALL	-s---	ic	l	nb	p-op	-	buffer address+content 7), 14)
corresponding MPI_WAIT (for BSEND req.)	---c-	c	l		p-op	-	buffer address 3), 7), 9)
corresponding MPI_WAIT (for other request)	---c-	c	nl		p-op	-	buffer address 7), 9)
corr. MPI_TEST returning flag=TRUE	---c-	c	l		p-op	-	buffer address 7), 9)
corr. MPI_TEST returning flag=FALSE	-----		l		p-op	-	buffer content+address 7)
corr. MPI_REQUEST_FREE (for inactive req-handle)	-----f	f	l		p-op	-	
MPI_CANCEL of nonblock./persistent pt-to-pt			l		p-op	-	
MPI_SENDRECV(REPLACE)	i-s-c-f	c+f	nl	b	b-op	-	